ML Domain Recruitment Task

Overview

This task is designed to evaluate your ability to work with real-world data the way it actually arrives: messy, incomplete, and without a roadmap. There are no step-by-step instructions for how to approach each problem. You are expected to make decisions, justify them, and communicate your findings clearly.

What we are assessing is not whether you can produce a high-accuracy model. We are assessing whether you understand what your data looks like, why you made the choices you did, and what your results actually mean.

Read this document in full before you begin. Every detail here is deliberate.

Dataset

You will work with the UCI Heart Disease Dataset (Cleveland subset). This is a real clinical dataset collected from patients at the Cleveland Clinic Foundation. It contains 303 records and 14 attributes covering demographic information, clinical measurements, and a binary diagnosis outcome.

Dataset Access:

Link: https://www.kaggle.com/datasets/ineubytes/heart-disease-dataset

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Column Reference

Column Name	Type	Description
`age`	Numerical	Age of the patient in years
`sex`	Categorical	1 = male, 0 = female
`cp`	Categorical	Chest pain type (0–3)
`trestbps`	Numerical	Resting blood pressure (mm Hg) on admission
`chol`	Numerical	Serum cholesterol in mg/dl — regression target
`fbs`	Categorical	Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)
`restecg`	Categorical	Resting ECG results (0–2)
`thalach`	Numerical	Maximum heart rate achieved
`exang`	Categorical	Exercise-induced angina (1 = yes, 0 = no)
`oldpeak`	Numerical	ST depression induced by exercise relative to rest
`slope`	Categorical	Slope of peak exercise ST segment (0–2)
`ca`	Numerical	Number of major vessels coloured by fluoroscopy (0–3); contains missing values marked as '?'
`thal`	Categorical	Thalassemia type (3 = normal, 6 = fixed defect, 7 = reversible defect); contains '?'
`target`	Binary	Diagnosis of heart disease (0 = absent, 1 = present)

Note: chol (serum cholesterol) is your regression target for Phase 4.

Do not treat it as a feature after Phase 2.

Task Phases

The task is divided into four phases. Complete them in sequence. Each phase builds on the previous one.

Phase 1: Understanding the Data

Before writing a single line of cleaning or modelling code, your first obligation is to understand what you are working with. This phase is entirely about observation.

Load the dataset and perform the following:

- Print the shape of the Data Frame and confirm it matches the expected 303 rows and 14 columns.
- Print the data types of all columns. Identify which columns have been loaded with an incorrect type due to the '?' entries.
- Identify all columns with missing or invalid values. '?' characters are missing values in this dataset. Quantify exactly how many are present per column.
- Describe the statistical summary of all numerical columns. Look at the min, max, and mean values and note anything that seems clinically implausible.
- Understand the shape and spread of your data. Visualise the distributions, relationships between variables, and any patterns or anomalies you can identify. Use the plots that are most appropriate for the question you are trying to answer. You are not told which plots to use — your choice of visualisation is part of what is being evaluated.

For every visualisation you produce, include one to two sentences beneath it explaining what the plot is telling you. A plot without an interpretation will not be considered.

Deliverable:

A written summary (minimum 5 sentences) describing what you observed about the dataset before cleaning it. This should cover the shape of the data, the nature of missing values, the distributions you found noteworthy, and any relationships between variables that stood out.

Phase 2 : Data Cleaning

Now that you understand the data, clean it. Every decision you make in this phase must be accompanied by a justification. Do not clean anything silently.

Address the following issues:

- Convert '?' entries to null values and handle them appropriately. State whether you chose to drop or impute, and why.
- Cast all columns to their correct data types after handling nulls.

- Examine the numerical columns for values that are clinically impossible or statistically extreme. Address them and explain your reasoning.
- Check for and remove exact duplicate rows if any are present.
- Encode categorical variables into a form suitable for a linear model. Explain the encoding strategy you chose.

There is no single correct way to clean this dataset. You will be assessed on whether your choices are logical and clearly communicated, not on whether they match a reference solution.

Deliverable:

Print the shape of your Data Frame before and after cleaning. For each decision you made (imputation strategy, outlier treatment, encoding choice), write one or two sentences explaining why you made that choice.

Phase 3 Feature Engineering

Create at least one new feature that is derived from existing columns and that you believe will be meaningful in predicting serum cholesterol. Generic or arbitrary transformations will be noted.

Some directions to consider — you are not required to use these and may propose something different:

- A feature combining age and maximum heart rate to proxy cardiovascular load.
- A binary flag identifying patients with both high resting blood pressure and ST depression, as this combination may carry clinical significance.
- An age group or risk tier derived from clinical thresholds rather than arbitrary quantile bins.

After creating your feature, visualise how it relates to the target variable chol. The visualisation should support your reasoning for why the feature might be useful.

Deliverables:

The new column(s) added to your Data Frame, and a written explanation (two to four sentences) of what the feature represents and why you expect it to carry predictive value.

Phase 4 — Linear Regression & Interpretation

Train a Linear Regression model to predict serum cholesterol (chol) using the cleaned and engineered features. The target column must be excluded from the feature set.

You are expected to:

- Split the data into training and test sets. State the split ratio you used.
- Fit a Linear Regression model on the training set.
- Report the R^2 score and Mean Absolute Error (MAE) on the test set.
- Print the coefficient for each feature. Identify the three features with the largest absolute coefficient values.
- Interpret those three coefficients in plain language. What does each one say about the relationship between that feature and cholesterol? A statement such as 'the coefficient is 12.4 so it is important' is not an interpretation.

A low R^2 score is not a failure. Cholesterol is difficult to predict from this feature set alone. What matters is that your interpretation is honest about what the model can and cannot tell you.

Deliverables:

Model metrics (R^2 and MAE) printed clearly. A written paragraph of four to six sentences interpreting your top three coefficients in clinical or practical terms, and a brief honest assessment of the model's limitations given the dataset.

What We Look For

- Demonstrates understanding of data shape, distributions, missing values, and anomalies. Plot choices reflect genuine analytical intent.
- Handles null values, impossible entries, and encoding correctly. Decisions are explained, not just executed.
- Creates at least one new feature that is clearly motivated and logically derived from domain understanding.
- Trains a linear model correctly, reports metrics, and interprets coefficients in plain language without jargon.

A Note from the ML Domain

We are not looking for perfect scores. We are looking for people who are curious about their data, honest in their reasoning, and precise in what they communicate. If you spend ten minutes exploring a column before cleaning it, that will show. If you copy a pipeline from a notebook and run it without understanding it, that will also show.

The best submissions we have seen are the ones where the candidate clearly had a conversation with the data and the analysis had a understanding point of view. We look forward to seeing yours.

Submission Guidelines

1. Submit a single Jupyter Notebook (.ipynb) file. All code, outputs, visualisations, and written interpretations must be contained within the notebook.
2. Ensure all cells are executed and outputs are visible. A notebook that requires re-running to view outputs will not be accepted.
3. Name your file in the format: FirstName_LastName_MLTask.ipynb
4. Submit via the link shared with you on the recruitment portal. Do not email submissions.
5. Deadline: As communicated on the recruitment portal. Late submissions will not be reviewed.

You may use any Python libraries you are comfortable with. There is no restriction on the tools used, provided the work is your own.

Wishing You All The Best
ML Domain ADG